

Paso 1: Creación del Clúster

```

$ aws eksctl create cluster --name=k8s-lab-cluster --region=$AWS_REGION --nodegroup-name=standard-workers --node-type=t3.medium --nodes=2 --managed
2026-04-17 22:03:38 [i] eksctl version 0.225.0
2026-04-17 22:03:38 [i] using region us-east-1
2026-04-17 22:03:38 [i] setting availability zones to [us-east-1f us-east-1c]
2026-04-17 22:03:39 [i] subnets for us-east-1f - public:192.168.0.0/19 private:192.168.64.0/19
2026-04-17 22:03:39 [i] subnets for us-east-1c - public:192.168.32.0/19 private:192.168.96.0/19
2026-04-17 22:03:39 [i] nodegroup "standard-workers" will use "" (amazon linux2023) 34
2026-04-17 22:03:39 [i] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain cu
rent behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-04-17 22:03:39 [i] using Kubernetes version 1.34
2026-04-17 22:03:39 [i] creating EKS cluster "k8s-lab-cluster" in "us-east-1" region with managed nodes
2026-04-17 22:03:39 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-04-17 22:03:39 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=k8s-lab-cluster'
2026-04-17 22:03:39 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "k8s-lab-cluster" in "us-east-1"
2026-04-17 22:03:39 [i] Cloudwatch logging will not be enabled for cluster "k8s-lab-cluster" in "us-east-1"
2026-04-17 22:03:39 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types=STREAM-YOUR-LOG-TYPES-HERE (e.g. all) --region=us-east-1 --cluster=k8s-lab-cluster'
2026-04-17 22:03:39 [i] default addons vpc-cni, kube-proxy, coredns, metrics-server were not specified, will install them as EKS addons
2026-04-17 22:03:39 [i]
2 sequential tasks: { create cluster control plane "k8s-lab-cluster",
2 sequential sub-tasks: {
2 sequential sub-tasks: {
1 task: { create addons },
}, wait for control plane to become ready,
}, create managed nodegroup "standard-workers",
}
2026-04-17 22:03:39 [i] building cluster stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:03:39 [i] deploying stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:04:09 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:04:40 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:05:40 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:06:41 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:07:41 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:08:41 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:09:42 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:10:42 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:11:43 [i] waiting for CloudFormation stack "eksctl-k8s-lab-cluster-cluster"
2026-04-17 22:11:45 [i] recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM per
missions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and run 'eksctl update a
addon
2026-04-17 22:11:45 [i] creating addon: vpc-cni
2026-04-17 22:11:46 [i] successfully created addon: vpc-cni
2026-04-17 22:11:46 [i] creating addon: kube-proxy
2026-04-17 22:11:47 [i] successfully created addon: kube-proxy
2026-04-17 22:11:47 [i] creating addon: coredns
2026-04-17 22:11:47 [i] successfully created addon: coredns

```

eksctl-k8s-lab-cluster-cluster

Delete stack

Update stack ▾

Stack actions ▾

Create stack ▾

Stack info	Events	Resources	Outputs	Parameters	Template	Change sets	Git sync
Overview Stack ID arn:aws:cloudformation:us-east-1:654654478122:stack/eksctl-k8s-lab-cluster-cluster/3305bac0-3ad3-11f1-9dc9-0e0f35a386e9 Description EKS cluster (dedicated VPC: true, dedicated IAM: true) [created and managed by eksctl] Status CREATE_COMPLETE Detailed status - Status reason - Root stack - Parent stack - Created time 2026-04-17 22:03:39 UTC-0500 Updated time - Deleted time - Drift status NOT_CHECKED Last drift check time - Termination protection Activated IAM role -							

Paso 2: Instalación y Parche del Metrics Server

```

$ kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
Warning: resource serviceaccounts/metrics-server is missing the kubectlkubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources cre
ated declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
serviceaccount/metrics-server configured
Warning: resource clusterroles/system:metrics-server is missing the kubectlkubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources
created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
clusterrole.rbac.authorization.k8s.io/system:metrics-server configured
Warning: resource clusterrolebindings/metrics-server-auth-reader is missing the kubectlkubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on reso
urces created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
clusterrolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader configured
Warning: resource clusterrolebindings/metrics-server-system:auth-delegator is missing the kubectlkubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should onl
y be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
clusterrolebinding.rbac.authorization.k8s.io/metrics-server-system:auth-delegator configured
Warning: resource clusterrolebindings/system:metrics-server is missing the kubectlkubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on re
sources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server configured
Warning: resource services/metrics-server is missing the kubectlkubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created de
claratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
service/metrics-server configured
Warning: resource deployments/metrics-server is missing the kubectlkubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created
declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
Warning: resource apiservices/v1beta1.metrics.k8s.io is missing the kubectlkubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources
created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io configured
The Deployment "metrics-server" is invalid:
  spec.template.spec.containers[0].ports[1].name: Duplicate value: "https"
  spec.selector: Invalid value: "matchLabels": {"app.kubernetes.io/instance": "metrics-server", "app.kubernetes.io/name": "metrics-server", "k8s-app": "metrics-server"}: field is immutable
$ kubectl patch deployment/metrics-server -n kube-system --type=json \
-p='[{"op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--kubelet-insecure-tls"}]'
deployment.apps/metrics-server patched
$ kubectl patch deployment/metrics-server -n kube-system -p '[{"spec":{"template":{"metadata":{"labels":{"k8s-app":"metrics-server"}}}}}]'
deployment.apps/metrics-server patched

```

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl get nodes
NAME                                STATUS    ROLES    AGE     VERSION
ip-192-168-30-108.ec2.internal      Ready    <none>   7m27s   v1.34.6-eks-bbe087e
ip-192-168-55-212.ec2.internal      Ready    <none>   7m27s   v1.34.6-eks-bbe087e

```

Paso 3: ConfigMap (Interfaz Web Personalizada)

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ cat <<'EOF' > index.html
<!DOCTYPE html>
<html lang="es">
<head>
<meta charset="UTF-8">
<title>EKS Lab - AWS Juan Esteban</title>
<style>
body { background-color: #1e1e2e; color: #cdd6f4; font-family:
sans-serif; display: flex; flex-direction: column; align-items:
center; justify-content: center; height: 100vh; margin: 0; }
h1 { color: #89b4fa; font-size: 3rem; }
.status { margin-top: 30px; padding: 15px 30px; backgroundcolor:
#a6e3a1; color: #11111b; border-radius: 8px; font-weight: bold;
font-size: 1.5rem; }
</style>
</head>
<body>
<h1> EKS Cluster Operativo - juan esteban</h1>
<p>¡Laboratorio de orquestación y autoescalado completado con
éxito!</p>
<p>Tráfico balanceado por AWS Network Load Balancer.</p>
<div class="status">HPA Status: ONLINE</div>
</body>
</html>
EOF

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl create configmap custom-website --from-file=index.html
configmap/custom-website created

```

Paso 4: Despliegue de la Aplicación (Generación con --dry-run)

4.1

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl create deployment nginx-deployment --image=nginx:alpine --dry-run=client -o yaml > deployment.yaml

```

```
deployment.yaml X
C: > Users > ELITEBOOK > deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    labels:
5      app: nginx-deployment
6    name: nginx-deployment
7  spec:
8    replicas: 1
9    selector:
10     matchLabels:
11       app: nginx-deployment
12   template:
13     metadata:
14       labels:
15         app: nginx-deployment
16     spec:
17       containers:
18       - image: nginx:alpine
19         name: nginx
20         resources:
21           requests:
22             cpu: "100m"
23             memory: "128Mi"
24         volumeMounts:
25         - name: web-volume
26           mountPath: /usr/share/nginx/html
27       volumes:
28       - name: web-volume
29         configMap:
30           name: custom-website
```

4.2

```
ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl create service loadbalancer nginx-service --tcp=80:80 --dry-run=client -o yaml > service.yaml
```

```

apiVersion: v1
kind: Service
metadata:
  annotations:
    service.beta.kubernetes.io/aws-load-balancer-type: "nlb"
  name: nginx-service
spec:
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: nginx-deployment
  type: LoadBalancer

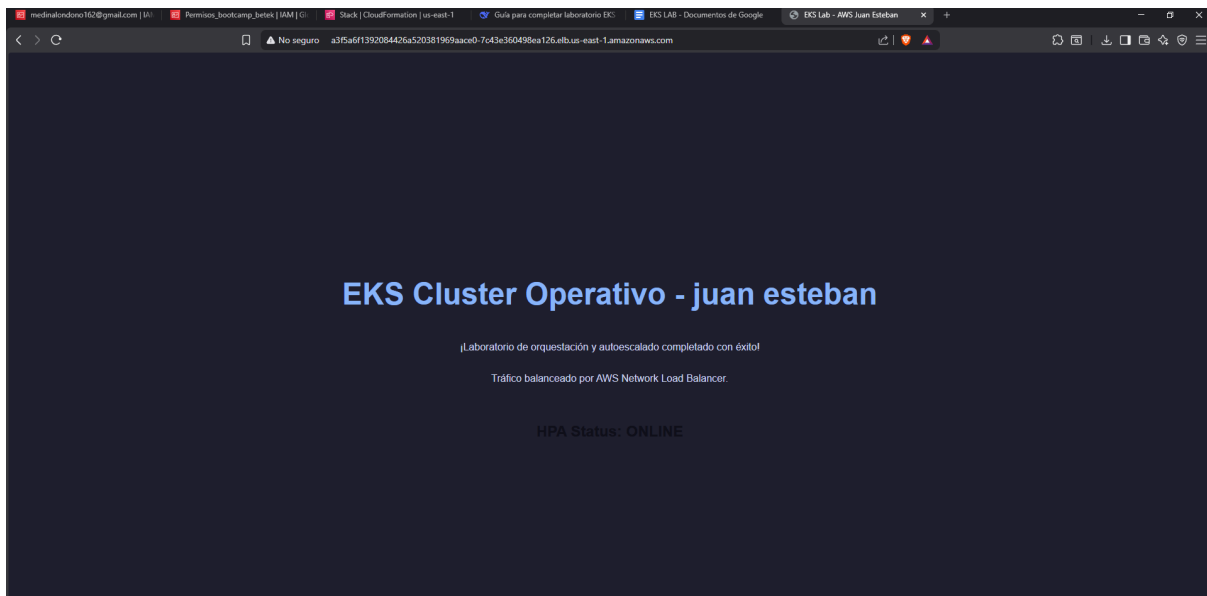
```

4.3

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl apply -f service.yaml
service/nginx-service created

```



Paso 5: Autoescalado (HPA) y Prueba de Estrés

5.1

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl autoscale deployment nginx-deployment --cpu-percent=30 --min=1 --max=5
Flag --cpu-percent has been deprecated, Use --cpu with percentage or resource quantity format (e.g., '70%' for utilization or '500m' for millicpu).
horizontalpodautoscaler.autoscaling/nginx-deployment autoscaled

```

5.2

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ cat <<'EOF' > stress.sh
#!/bin/bash
URL=$1
echo "Atacando $URL con 50 hilos..."
for i in $(seq 1 50); do
    while true; do
        curl -s -o /dev/null "$URL"
    done &
done
wait
EOF
ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ chmod +x stress.sh

```

5.3

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl get hpa -w
NAME                REFERENCE                TARGETS      MINPODS  MAXPODS  REPLICAS  AGE
nginx-deployment    Deployment/nginx-deployment  cpu: 17%/30%    1         5         1         3m43s
nginx-deployment    Deployment/nginx-deployment  cpu: 18%/30%    1         5         1         3m45s

```

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ ./stress.sh http://a3f5a6f1392084426a520381969aace0-7c43e360498ea126.elb.us-east-1.amazonaws.com/
Atacando http://a3f5a6f1392084426a520381969aace0-7c43e360498ea126.elb.us-east-1.amazonaws.com/ con 50 hilos...

```

Paso 6: Observabilidad con CloudWatch

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl top pods
NAME                                CPU(cores)  MEMORY(bytes)
nginx-deployment-ddd7f5796-mhj4q    22m         3Mi

```

```

ELITEBOOK@DESKTOP-GN39F7E MINGW64 ~
$ kubectl get hpa
NAME                REFERENCE                TARGETS      MINPODS  MAXPODS  REPLICAS  AGE
nginx-deployment    Deployment/nginx-deployment  cpu: 23%/30%    1         5         1         10m

```

preguntas repaso:

1. ¿Cuál es el beneficio principal de inyectar una página web a través de un ConfigMap en lugar de compilarla dentro de la imagen Docker?

R/= El beneficio principal es el desacople entre el código y la configuración

2. ¿Por qué el HPA requiere estrictamente que se definan los requests de CPU en el contenedor?

R/= El HPA utiliza la métrica de utilización para decidir cuándo escalar.

3. En la prueba de estrés, ¿por qué fue necesario bajar el averageUtilization al 30% para forzar el escalado?

R/= Si el escalado no ocurre al umbral predeterminado, es probable que se deba a la inercia de las métricas o a un desajuste en el muestreo.

4. ¿Qué función cumple la anotación aws-load-balancer-type: "nlb" en el Servicio?

R/= Esta anotación instruye al AWS Load Balancer Controller para que, en lugar de aprovisionar un CLB, provisione un NLB.

5. Si el API Server reporta MissingEndpoints para un Servicio, ¿cuál es la causa más probable?

R/= La causa número uno suele ser un desajuste entre los Selectores del Servicio y las Labels de los Pods.